



Sonu Gupta

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Professional Summary

A highly analytical and forward-thinking Computer Engineering graduate specializing in backend development, AI integrations, and full-stack system architectures. Demonstrates a strong technical track record in configuring robust relational data configurations, machine learning inference flows, and optimized pipeline automation. Experienced in cross-functional leadership, design-to-code conversions, and deploying user-centric digital applications. Passionate about resolving algorithmic infrastructure bottlenecks and writing clean, scalable codebase ecosystems.

Education

TRIBHUVAN UNIVERSITY

Bachelor in Computer Engineering

Seventh Semester Score: 73.58% Coursework: Database Management Systems, Software Engineering, Computer Networks, Data Analysis.

IOE, Purwanchal Campus, Dharan, Nepal

Result Awaiting (Exams Completed, 2026)

LORD BUDDHA HIGHER SECONDARY SCHOOL

National Examination Board +2, Science, 2021

Biratnagar, Nepal

Class 12 Score: 3.19 GPA

CORBAN ENGLISH SCHOOL

Secondary Education Exam (SEE / SLC), 2019

Biratnagar, Nepal

Aggregate Score: 3.30 GPA

Technical Skills & Core Competencies

Languages & Frameworks:	Python, JavaScript (ES6+), React.js, Node.js, Express.js, FastAPI, SQL, Tailwind CSS.
Data & Cloud Infrastructure:	Relational Databases (SQL, MongoDB), Vector DBs, System Automation, CMS (WordPress).
Specialized Domains:	Advanced Data Pipelines, Speech Processing, Machine Learning, Information Retrieval (RAG).
Core Methodologies:	Full-stack platform mapping, cross-functional leadership, structured technical documentation.

Professional Experience & Leadership

2025–2026	Google Gemini Student Ambassador Acted as the primary community touchpoint bridging complex digital technologies with the campus student base. Spearheaded large-scale technical bootcamps, mentored junior development cohorts, and delivered presentations on artificial intelligence frameworks, cloud configurations, and technical solution building.	IOE Purwanchal Campus
2022–Present	Frontend Developer & Technical Consultant Engineered and deployed over 10 fully responsive, production-ready web platforms using modern React.js architectures and content engines. Transformed complex UI/UX drafts into modular, high-speed frontend blocks, maintaining full back-end API integration and absolute code stability.	Remote / Freelance

Research & Publications Tracking

Ongoing	Data Pipeline Architecture for Machine Learning Systems Conducting engineering research targeting the design of modular pipeline architectures optimized for heavy machine learning model training flows. Investigating methodologies for automated file ETL execution, data structural compression, and removing pipeline processing bottlenecks.
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Technical Engineering AI Systems Portfolio

2025–2026	AI-Powered Visual Search Platform (e-Pasal) Processed and indexed an exhaustive big-data cluster exceeding 100,000 images, fine-tuning deep feature embeddings utilizing Vision Transformer (ViT) designs. Built out high-performance FastAPI server lines for real-time visual product lookup and cross-platform stock SKU matching.
2025	Multilingual RAG Question Answering System Constructed a Retrieval-Augmented Generation (RAG) informational layout engineered for regional document querying supporting localized Nepali and Hindi languages. Integrated vectorized embedding lookups alongside neural transformers to secure real-time contextual question resolutions.
2024	Stutter Detection System for Regional Languages Designed complex speech preprocessing matrices and architectural feature selection modules to evaluate and dissect unique phonetic patterns in South Asian speech. Configured neural classification parameters to dynamically audit audio signal integrity.
2023	Scalable E-Commerce & Admin Dashboard Engine Designed a fully comprehensive software commerce framework utilizing full-stack JS logic, complete with protected user authentication, product logs, cart calculations, and a complete administrative command dashboard. Optimized schema indexes to secure sub-second response limits.

Academic Conferences & Professional Attributes

- **Conferences & Exposure:** Attended the *International Conference on Recent Trends in Artificial Intelligence (ICRTAI)*, Birtamode, 2025, tracking developments in scalable cloud infrastructure and localized transformer networks.
- **System Optimization Focus:** Exceptionally precise troubleshooting approach natively suited for code refactoring, database load distribution, system testing, and logic debugging.
- **Cross-Functional Communication:** Proven interpersonal, collaboration, and translation abilities honed through technical freelance consulting and elite global student leadership positions.